UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

INTEGRATED DNA TECHNOLOGIES, INC.,
Petitioner,

v.

TECAN GENOMICS, INC.,
Patent Owner.

IPR2024-01504
Patent 11,098,357 B2

Before GEORGIANNA W. BRADEN, ZHENYU YANG, and TIMOTHY G. MAJORS, *Administrative Patent Judges.*

BRADEN, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

In this *inter partes* review, Integrated DNA Technologies, Inc. ("Petitioner") challenges the patentability of claims 1–4, 6, 7, 9, and 10 of U.S. Patent No. 11,098,357 B2 (Ex. 1004, "the '357 patent"), which is assigned to Tecan Genomics, Inc. ("Patent Owner").

We have jurisdiction under 35 U.S.C. § 6. This Final Written Decision, issued pursuant to 35 U.S.C. § 318(a), addresses issues and arguments raised during the trial in this *inter partes* review. For the reasons discussed below, we determine Petitioner has proven by a preponderance of the evidence that claims 1–4, 6, 7, 9, and 10 of the '357 patent are unpatentable in view of the challenges based on Bielas alone as well as the combined teachings of Iafrate and Kivioja. *See* 35 U.S.C. § 316(e) (2018) ("In an inter partes review instituted under this chapter, the petitioner shall have the burden of proving a proposition of unpatentability by a preponderance of the evidence.").

A. Procedural History

Integrated DNA Technologies, Inc. filed a Petition (Paper 1, "Pet.") challenging claims 1–4, 6, 7, 9 and 10 of the '357 patent on the following basis:

Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1–4, 6, 7, 9, 10	103	Iafrate,[1] Kivioja[2]
1–4, 6, 7, 10	102	Bielas[3]
1–4, 6, 7, 9, 10	103	Bielas

Pet. 26. In support of its patentability challenge, Petitioner relies on, *inter alia*, the Declaration of Peter A. Sims, Ph.D. ("Dr. Sims"). Ex. 1079. Patent Owner filed a Preliminary Response. Paper 6 ("Prelim. Resp."). Trial was instituted on the asserted challenges to patentability. Paper 7 ("Inst. Dec."), 42.

During the trial, Patent Owner filed a Response (Paper 21, "PO Resp.") to the Petition. In support of its Response, Patent Owner relies on, *inter alia*, the Declaration of Carlos D. Bustamante, Ph.D. ("Dr. Bustamante"). Ex. 2035. Petitioner filed a Reply to Patent Owner's Response (Paper 29, "Pet. Reply"), and Patent Owner filed a Sur-reply to Petitioner's Reply (Paper 34, "PO Sur-reply").

An oral hearing was held on January 22, 2026, a transcript of which appears in the record. Paper 39 ("Tr.").

B. Real Parties-in-Interest

Petitioner identifies "Integrated DNA Technologies, Inc., and its corporate parent, Danaher Corporation," as real parties-in-interest. Pet. 73.

[1] U.S. Patent Application Publication No. 2013/0303461 A1, published November 14, 2013 (Ex. 1011) ("Iafrate").
[2] Kivioja et al., *Counting absolute numbers of molecules using unique molecular identifiers*, 9(1) Nature Methods 72 (dated Jan. 2012 and published online Nov. 20, 2011) (Ex. 1012) ("Kivioja").
[3] International Patent Application Publication No. WO 2013/123442 A1, published August 22, 2013 (Ex. 1013) ("Bielas").

Patent Owner identifies "Tecan Group AG (also known as Tecan Group Ltd.)" as a real-party-in-interest. Paper 4 (Patent Owner's Mandatory Notices), 1; Paper 26 (Patent Owner's Updated Mandatory Notices), 1.

C. Related Matters

The parties indicate that the '357 patent has been asserted in the following district court litigation: *Tecan Genomics, Inc. v. Invitae Corp. et al*, C.A. No. 23-cv-1114-GBW (D. Del.). Pet. 73; Paper 4, 1; Paper 26, 1. The parties also cites *Tecan Genomics, Inc. v. QIAGEN Sciences, LLC*, C.A. No. 23-cv-1115-GBW (D. Del.) as a related matter because it involves patents in the same family as the '357 patent. Pet. 73; Paper 26, 1.

The parties indicate that Petitioner submitted petitions for *inter partes* review of U.S. Patent No. 9,546,399 in IPR2024-01502 and U.S. Patent No. 11,725,241 in IPR2024-01506. Pet. 73; Paper 4, 1. Patent Owner also states that Petitioner submitted petitions for *inter partes* review of U.S. Patent No. 10,036,012 in IPR2025-00015 and for U.S. Patent No. 10,876,108 in IPR2025-00016. Paper 4, 1.

Patent Owner also states that QIAGEN Sciences, LLC, submitted a Petition for *inter partes* review of the '357 patent (IPR2025-00026), the '241 patent (IPR2025-00027), the '012 patent (IPR2025-00028), and the '108 patent (IPR2025-00029). *Id.*

D. The '357 Patent

The '357 patent is titled "Compositions and Methods for Identification of a Duplicate Sequencing Read" and issued on August 24, 2021. Ex. 1004, codes (45), (54). It issued from an application filed January 15, 2020 and claims priority to a provisional application filed on November 13, 2013. *Id.* at codes (21), (22), (60).

1. Written Description

The ’357 patent “relates generally to the field of high throughput sequencing reactions and the ability to discern artifacts arising through sequence duplications from nucleotide molecules that are unique molecules.” *Id.* at 1:27–30. The ’357 patent explains that PCR duplicate artifacts may occur in RNA sequencing applications during library amplification, which hampers accurate gene expression measurements. *Id.* at 1:35–37. The ’357 patent also states that paired end sequencing adds cost in expression analysis studies. *See id.* at 1:56–62. In view of this, the ’357 patent expresses a need for improved methods of “low-cost, high through-put sequencing of regions of interest, genotyping or simple detection of RNA transcripts without inherent instrument inefficiencies that drive up sequencing costs due to the generation of unusable or non-desired data reads.” *Id.* at 1:65–2:3.

The ’357 patent states that it is based on methods for discerning “duplicate sequencing reads from a population of sequencing reads.” *Id.* at 2:15–17. The method includes “ligating an adaptor to a 5′ end of each nucleic acid fragment of a plurality of nucleic acid fragments from one or more samples, wherein the adaptor comprises an indexing primer binding site, an indexing site, an identifier site, and a target sequence primer binding site.” *Id.* at 2:24–29. The ’357 patent describes an embodiment in which “the identifier site is between 1 and 8 nucleotides in length.” *Id.* at 2:62–64. The ’357 patent explains that “[t]he ligated adaptor-nucleic acid fragment products can be amplified, thus generating a population of sequencing reads from the amplified adaptor-nucleic acid ligation products.” *Id.* at 2:29–32. In addition, “[t]he sequencing reads with a duplicate identifier site and target sequence can then be detected from the population of sequencing reads." *Id.* at 2:32–34.

2. Illustrative Claim

Claim 1, reproduced below, is the sole independent challenged claim and is illustrative of the subject matter:

> 1. A method for detecting duplicate sequencing reads, the method comprising:
>
> ligating an adaptor to each of a plurality of nucleic acid fragments, wherein each adaptor comprises a unique identifier having from about 1 to about 8 nucleotides; an indexing site unique to a subset of the adaptors, and a primer binding site;
>
> amplifying the adaptor-ligated fragments into amplicons;
>
> sequencing the amplicons to produce sequence reads that include identifier and target sequences; and
>
> identifying sequence reads with identical identifier and target sequences as duplicates.

Ex. 1001, 25:36–47.

II. PRELIMINARY MATTERS

A. Legal Standards

A "prior art reference—in order to anticipate under 35 U.S.C. § 102—must not only disclose all elements of the claim within the four corners of the document, but must also disclose those elements 'arranged as in the claim.'" *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1369 (Fed. Cir. 2008) (quoting *Connell v. Sears, Roebuck & Co.*, 722 F.2d 1542, 1548 (Fed. Cir. 1983)). "A single prior art reference may anticipate without disclosing a feature of the claimed invention if such feature is necessarily present, or inherent, in that reference." *Allergan, Inc. v. Apotex Inc.*, 754 F.3d 952, 958 (Fed. Cir. 2014).

"[A]nticipation by inherent disclosure is appropriate only when the [single prior art] reference discloses prior art that must *necessarily* include the unstated limitation." *Transclean Corp. v. Bridgewood Servs., Inc.*, 290 F.3d 1364, 1373 (Fed. Cir. 2002) (citation omitted). "Inherency . . . may not be established by probabilities or possibilities. The mere fact that a certain thing *may* result from a given set of circumstances is not sufficient." *Cont'l Can Co. USA, Inc. v. Monsanto Co.*, 948 F.2d 1264, 1269 (Fed. Cir. 1991) (internal quotation marks and citations omitted). Rather, "[t]he inherent result must inevitably result from the disclosed steps." *In re Montgomery*, 677 F.3d 1375, 1380 (Fed. Cir. 2012). Of course, anticipation "is not an 'ipsissimis verbis' test." *In re Bond*, 910 F.2d 831, 832–33 (Fed. Cir. 1990) (citing *Akzo N.V. v. United States Int'l Trade Comm'n*, 808 F.2d 1471, 1479 & n.11 (Fed. Cir. 1986)). "An anticipatory reference . . . need not duplicate word for word what is in the claims." *Standard Havens Prods. v. Gencor Indus.*, 953 F.2d 1360, 1369 (Fed. Cir. 1991).

A patent claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. 35 U.S.C. § 103; *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). "[W]hen a patent claims a structure already known in the prior art that is altered by the mere substitution of one element for another known in the field, the combination must do more than yield a predictable result." *KSR*, 550 U.S. at 416 (citing *United States v. Adams*, 383 U.S. 39, 50–51 (1966)). The question of obviousness is resolved based on underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of non-obviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

In an *inter partes* review, a petition must identify "with particularity, each claim challenged, the grounds on which the challenge to each claim is based, and the evidence that supports the grounds for the challenge to each claim." 35 U.S.C. § 312(a)(3) (2018); *see also* 37 C.F.R. § 42.104(b) (2024) (requiring a petition for *inter partes* review to identify how the challenged claim is to be construed and where each element of the claim is found in the prior art patents or printed publications relied upon).

"In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3)); *see also Intelligent Bio-Sys.*, 821 F.3d at 1369. This burden of persuasion never shifts to Patent Owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in *inter partes* review). Therefore, to prevail in an *inter partes* review, Petitioner must explain how the proposed combination of prior art would have rendered the challenged claims unpatentable. At this final stage, we determine whether the information presented in the Petition demonstrates by a preponderance of the evidence that the challenged claims would have been obvious over the asserted prior art.

We analyze the challenges presented in the Petition in accordance with the above-stated principles.

B. Level of Ordinary Skill in the Art

In determining the level of ordinary skill in the art, various factors may be considered, including the "type of problems encountered in the art;

prior art solutions to those problems; rapidity with which innovations are made; sophistication of the technology; and educational level of active workers in the field." *In re GPAC, Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995) (quoting *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962, (Fed. Cir. 1986)). Furthermore, the prior art itself can reflect the appropriate level of ordinary skill in the art. *Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

Here, Petitioner asserts that "[b]ecause the '357 patent is directed to library preparation and data interpretation methods for NGS [next generation sequencing], the POSA could have had academic training in any of a variety of fields that used NGS, including chemistry, molecular biology, biochemistry, or medicine." Pet. 21 (citing Ex. 1079 ¶¶ 184–186, 190). According to Petitioner, one of ordinary skill in the art "would have a PhD or equivalent training in one of those fields and several years of work experience (or commensurately less education and more work experience)." *Id.* (citing Ex. 1079 ¶¶ 184–186). Petitioner contends that a person of ordinary skill in the art "further would have had specific experience with implementing and designing library preparation methods for Illumina, the leading NGS platform in 2013" and "[t]his would have included specific experience with, and understanding of, the sequences contained in adaptors for Illumina sequencing at the priority date, and the chemistry used to attach those adaptors to DNA fragments prior to sequencing." *Id.* at 21–22 (citing Ex. 1079 ¶¶ 184–186).

Patent Owner initially disputed the experience level Petitioner asserts for one of ordinary skill in the art. Prelim. Resp. 8–9. Specifically, Patent Owner argued that "[t]his level of skill inappropriately requires the POSA to be an *inventor* of library preparation methods, which is far above 'ordinary skill.'" *Id.* at 8. According to Patent Owner, one of ordinary skill "would have had experience with DNA sequencing methods and a working knowledge of how these sequencing methods functioned." *Id.*

On the initial trial record, we agreed with Patent Owner that Petitioner's asserted level of ordinary skill in the art is higher than warranted when Petitioner argues that a person of ordinary skill "further would have had specific experience with implementing and designing library preparation methods." Pet. 21–22. We disagreed with Patent Owner, however, that an ordinarily skilled artisan only needs to "have had experience with DNA sequencing methods" (*see* Prelim. Resp. 8) because an artisan with experience based only on pre-NGS sequencing methods would not qualify as a person of ordinary skill.

Instead, we found that a person of ordinary skill would have had either (1) a Ph.D. in the field of chemistry, molecular biology, or biochemistry or (2) a B.S. or M.S. degree with three to five years of experience working in any of those fields. Inst. Dec. 9. We specifically found that "[t]he person of ordinary skill would have had experience with NGS sequencing methods and a working knowledge of how those methods functioned." *Id.*

Patent Owner now submits that it does not dispute the definition of a person of ordinary skill in the art as laid out in the Institution Decision. PO Resp. 7–8.

Reviewing the entire trial record, we maintain the same definition of a person of ordinary skill in the art and note that it is consistent with the skill level reflected in the disclosures of the '357 patent and prior art. We further note that the skill level does not affect our analysis or our final decision.

C. Claim Construction

Claims are construed "in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent." 37 C.F.R. § 42.100(b) (2024). Under this standard, claim terms are generally given their plain and ordinary meaning as would have been understood by a person of ordinary skill in the art at the time of the invention and in the context of the entire patent disclosure. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1313 (Fed. Cir. 2005) (*en banc*). In some cases, the patentee may serve as its own lexicographer. *Grace Instrument Industries, LLC v. Chandler Instruments Co. LLC*, 57 F.4th 1001, 1010 (Fed. Cir. 2023) (*quoting Phillips*, 415 F.3d at 1316).

The following terms were either identified by the parties (Pet. 22–23; PO Resp. 9–19) or the Board for construction (Inst. Dec. 10–13).

1. "indexing site"

Petitioner argues that the '357 patent defines "indexing site" as a sample index, specifically, "a nucleic acid sequence that acts as an index for multiple polynucleotide samples, thus allowing for the samples to be pooled together into a single sequencing run, which is known as multiplexing." Pet. 23 (citing Ex. 1004, 4:40–44). Patent Owner did not disagree initially with Petitioner's proposed construction other than to argue that no claim requires construction. Prelim. Resp. 9.

In our Institution Decision, we noted that the '357 patent states:

> An indexing site is a nucleic acid sequence that acts as an index for multiple polynucleotide samples, thus allowing for the samples to be pooled together into a single sequencing run, which is known as multiplexing.

Ex. 1004, 4:40–44. We understood the above-quoted passage of the ’357 patent to provide a definition for “indexing site,” and we construed the term in accordance therewith. The parties do not dispute this definition and we maintain that construction for our analysis here. *See* PO Resp. 9; Reply 6–14.

2. “identifier site”

Petitioner argues that the ’357 patent defines “identifier site” as a molecular tag attached to fragments prior to amplification, specifically, “a nucleic acid sequence that comprises random bases and is used to identify duplicate sequencing reads.” Pet. 22 (citing Ex. 1004, 4:49–51). Petitioner argues that an “identifier site” “can be designed as a set of sequences, or it can be semi-random, or it can be completely random.” *Id.* (citing Ex. 1004, 4:55–57). Petitioner argues that an “identifier site” can also “be a fixed length, or it can be a variable length.” *Id.* at 22–23 (citing Ex. 1004, 4:57–58).

Patent Owner did not disagree initially with Petitioner’s proposed construction other than to argue that no claim requires construction. Prelim. Resp. 9.

In our Institution Decision, we noted that the ’357 patent states: “An identifier site is a nucleic acid sequence that comprises random bases and is used to identify duplicate sequencing reads.” Inst. Dec. (quoting Ex. 1004, 4:49–51). We understood this passage of the ’357 patent to provide a definition for “identifier site,” and we construe the term in accordance therewith. The parties do not dispute this definition and we maintain that construction for our analysis here. *See* PO Resp. 9; Reply 6–14.

3. "identifying sequence reads with identical identifier and target sequences as duplicates"

The parties did not submit constructions of this limitation initially. Rather, the Board requested that the parties address the construction of this limitation after institution. Inst. Dec. 10–13; *see Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999) (claims are construed only to the extent necessary to resolve a dispute).

Patent Owner contends the limitation should be construed as "identifying sequence reads as duplicates through a step that uses both the identifier and at least a portion of the target sequence." PO Resp. 9; Sur-Reply 2. Patent Owner argues that Example 5 of the '357 patent supports its interpretation because it describes a process of identifying duplicate sequencing reads using both the identifier and a portion of the target sequence. PO Resp. 11 (citing Ex. 1004, 20:19–21; Ex. 2035 ¶¶ 85–88). Patent Owner further argues that the "target sequence read" includes "***both*** the claimed identifier sequence and a portion of the target sequence." *Id.* at 12 (citing Ex. 2035 ¶¶ 85–88). Patent Owner contends the limitation does not require the use of entire "target sequence" because that is inconsistent with both the Specification and the prosecution history of the '357 patent. Sur-Reply 2. Patent Owner further explains that the "target sequence" and a sequence "read" may not be coextensive because the read may vary in length and can be shorter than the entire target sequence. Tr. 38:26–39:14. Patent Owner concedes, however, that if the "target sequence" and the sequence read are coextensive, then there is no dispute with Petitioner's position. *Id.*

Patent Owner then relies on the prosecution history of related Patent Application No. 14/540,917 to indicate that the "identify" step of the limitation is an ***active*** step. PO Resp. 14 (citing Ex. 1018, 797–781).

According to Patent Owner, an ordinarily skilled artisan would have understood that "Applicants were directing the claims at a method that includes a step in which duplicates are identified using a combination of the identifier site and at least a portion of the target sequence." *Id.* at 15 (citing Ex. 1019, 453). Patent Owner, therefore, concludes the claims require using both the information from the identifier and the portion of the target sequence together for "identifying duplicates" in the same step. Sur-Reply 2–8 (Ex. 1004, 18–20; Ex. 1018, 797–99; Ex. 1019, 453; Ex. 1124, 51:15–53:11, 86:24–88:11–20).

Petitioner contends that the plain language of the claim and the Specification require the "duplicates" to be "sequence reads with identical identifier and target sequences." Reply. 6 (citing Ex. 1004, 25:46–47). According to Petitioner, this means that the "duplicate" to be "100% identical, i.e., having a Hamming distance of zero because this is the plain and ordinary meaning of "identical" as used in the claim. *Id.* (citing Ex. 1124, 37:24–38:3). Petitioner asserts that Patent Owner's interpretation inappropriately rewrites the claim to require a temporal limitation (a step) and allow sequence comparison of less than the entirety of the target sequence ("at least a portion of"). *Id.* at 7–13; Tr. 10:20–17:13. Petitioner specifically argues the claims only require identifying sequence reads with identical identifier and target sequences, stating "[y]our two data inputs use those to identify duplicates. Nowhere in the claim is there a limitation that dictates when in the process of identifying duplicates that you need to use those two data inputs. [Patent Owner's] position, on the other hand, imposes simultaneity . . . ." Tr. 11:7–12.

Yet Patent Owner states that the claim does not require the use of the identifier and target sequence simultaneously or even that there is a temporal requirement where the identifier needs to be compared before the target sequence. Tr. 34:23–35:12. Rather, Patent Owner appears to argue that as long as the limitations are performed as some kind of "step," then the method claims are satisfied. *Id.*

We observe that the '357 patent states: "[a] duplicate sequencing read can be a sequencing read with the same identifier site and target sequence as another sequencing read in the population of sequencing reads." Ex. 1004, 4:11–15. The claim does not use the exact wording as the Specification.

The '357 patent states that, when analyzing RNA sequence data, and when two or more identical sequences are found, "it can be difficult to know if these represent unique cDNA molecules derived independently from different RNA molecules, or if they are PCR duplicates derived from a single RNA molecule." Ex. 1004, 1:39–42.

The '357 patent distinguishes between "apparent duplicates" and "true duplicates" as follows:

> In some embodiments, duplicate sequencing reads are differentiated from one another as being true duplicates versus apparent or perceived duplicates. Apparent or perceived duplicates may be identified from sequencing libraries and using conventional measures of duplicate reads (i.e., reads were mapped using bowtie), where all reads with the same start and end nucleic acid coordinates were counted as duplicates. True duplicates may be identified from sequencing libraries having had an identifier site introduced through ligation to differentiate between fragments of DNA that randomly have the same start and end mapping coordinates.

Ex. 1004, 14:21–31; *see id.* at Fig. 5; *see also* Example 1, 18:6-15 (When evaluating two million random reads, the apparent duplicates were found to comprise 39% of all reads. However, the true duplicates, identified through the use of an identifier site, were found to comprise only 26% of all reads. The methods employing the use of an identifier site were found to considerably increase the resolution of the true number of duplicates within the pool of reads.). The '357 patent also discloses in Example 5: "Any read that has identical sequence to any other read is identified as a duplicate and removed from the population, thus, only a single copy is retained within the file." Ex. 1004, 20:60–62.

Analyzing the patent disclosure in light of the parties' arguments, we determine that "duplicates" in claim 1 of the '357 patent means the sequences possess an identical sequence for both the identifier and the length of the sequence read that is obtained. Additionally, we agree with the parties that there is no temporal requirement for "identifying sequence reads" using the identifier and the target sequence, and the two do not have to be used simultaneously. Rather, based on both Examples 1 and 5 of the '357 patent, the claims only require that the two data points be used for "identifying sequence reads."

4. "comprising"

The Board preliminarily construed "comprising" as an open-ended transition that is not limited to the listed steps. *See* Inst. Dec. 13 (citing *Vivid Techs., Inc.*, 200 F.3d at 811). Patent Owner agrees that the "comprising" transitional phrase of the challenged claims leaves open the possibility that additional steps can be included in the method, but argues that it should not "be interpreted so broadly as to effectively eliminate the step of identifying duplicates as claimed." PO Resp. 18. Patent Owner argues that a "broad interpretation of the claims cannot be read to include a method in which the identifying duplicates step only identifies sequences with PCR or SBS errors and does not identify duplicates in the manner claimed." *Id.* According to Patent Owner, "[e]xpanding the claims to include only identification of reads without duplicate sequences (those that contain PCR or SBS errors) makes this limitation superfluous and cannot be the correct interpretation." *Id.* Patent Owner argues that the proper construction of the claims (1) covers a method of identifying sequence reads that are identical as duplicates through a process that uses both the identifier and at least a portion of the target sequence, and (2) does not exclude methods that include a step of identifying PCR errors in addition to the claimed step of identifying duplicates. *Id.* at 19.

Petitioner agrees with Patent Owner and states "[a]ll claim 1 requires is that two sequences are identified as duplicates using the claimed method, even if the method identifies as duplicates additional sequences that are not 100% identical." Reply 14 (citing Ex. 1124, 55:11–57:3; Ex. 1117).

The Board agrees with the parties and we construe "comprising" as an open-ended transition, meaning the claims require, but are not limited to, the listed steps.

III. ANALYSIS

A. Alleged Obviousness of Claims 1–4, 6, 7, 9, and 10 over Iafrate and Kivioja

Petitioner contends that claims 1–4, 6, 7, 9, and 10 would have been obvious over Iafrate and Kivioja. Pet. 24–47. Patent Owner disagrees. *See* PO Resp. 24–33.

1. Summary of the Prior Art

a) Iafrate

Iafrate is a U.S. Patent Application Publication titled "Methods for Determining a Nucleotide Sequence." Ex. 1011, code (54). More specifically, Iafrate is "directed to methods of determining oligonucleotide sequences, e.g. by enriching target sequences prior to sequencing the sequences." *Id.* at code (57).

Iafrate describes a method that includes:

> (a) ligating a target nucleic acid comprising the known target nucleotide sequence with a universal oligonucleotide tail-adaptor; (b) amplifying a portion of the target nucleic acid and the amplification strand of the universal oligonucleotide tail-adaptor with a first adaptor primer and a first target-specific primer; (c) amplifying a portion of the amplicon resulting from step (b) with a second adaptor primer and a second target-specific primer; [and] (d) sequencing the amplified portion from step (c) using a first and second sequencing primer.

Id. ¶ 7. Iafrate also discloses embodiments in which "the universal oligonucleotide tail-adaptor can further comprise a barcode portion." *Id.* ¶ 13.

Iafrate's Figure 9, which shows "schematics of illustrative universal oligonucleotide tail-adaptors (SEQ ID NOS 45-48, respectively, in order of appearance)," is reproduced below. *Id.* ¶ 33.

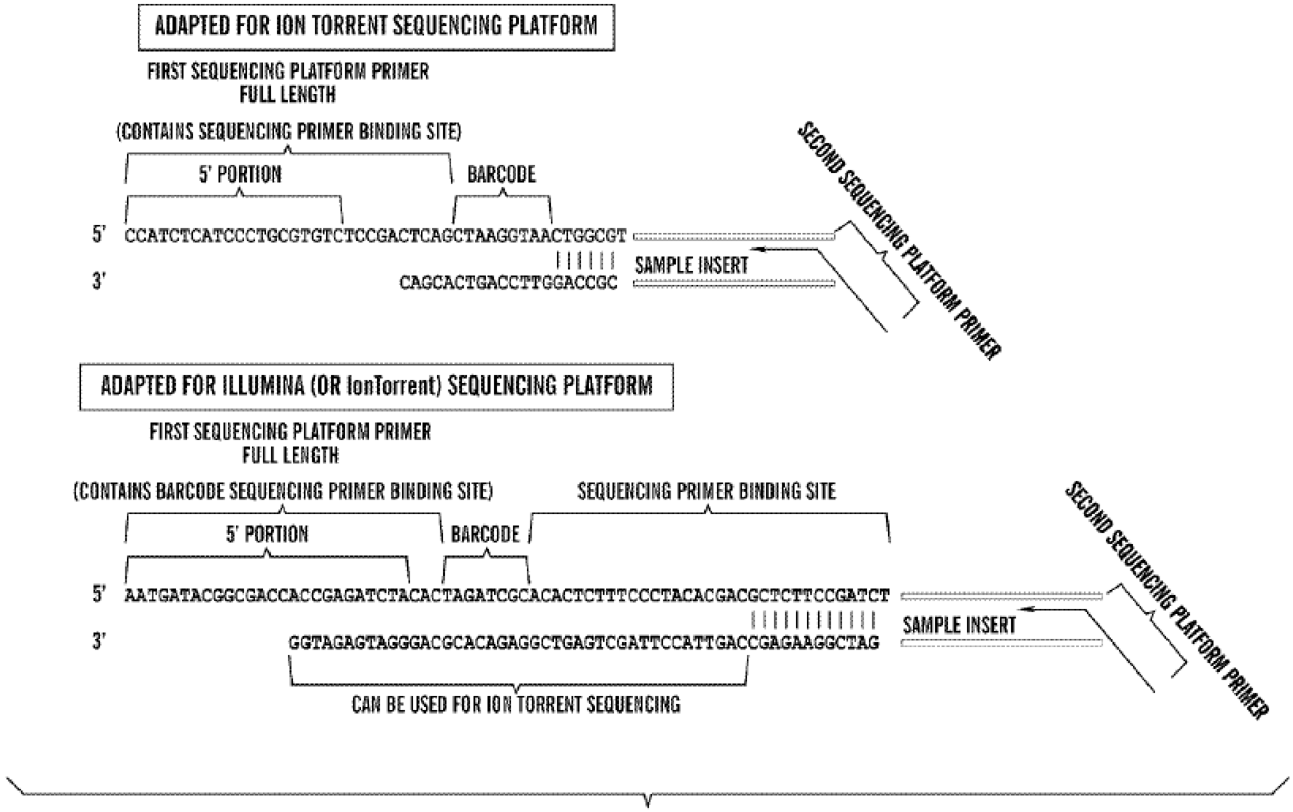


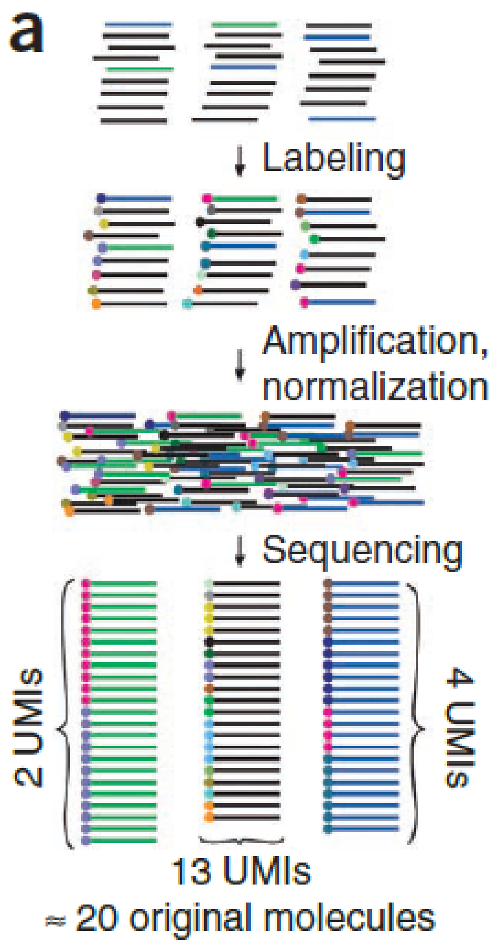
FIG. 9

b) Kivioja (Ex. 1012)

Kivioja is a journal article titled "Counting absolute numbers of molecules using unique molecular identifiers." Ex. 1012, 1. Kivioja explains that "[c]ounting individual RNA or DNA molecules is difficult because they are hard to copy quantitatively for detection." *Id.* In view of this, Kivioja describes applying "unique molecular identifiers (UMIs), which make each molecule in a population distinct, to genome-scale human karyotyping and mRNA sequencing in *Drosophila melanogaster*." *Id.* According to Kivioja, "this method can improve accuracy of almost any next-generation sequencing method, including chromatin immunoprecipitation–sequencing, genome assembly, diagnostics and manufacturing-process control and monitoring." *Id.*

In Kivioja's method, "each molecule in a population is first made unique," which "can be accomplished by adding a random DNA sequence label, by fragmenting or by taking an aliquot of a complex mixture that is small enough to contain only distinct molecules." *Id.* According to Kivioja, "[a]ny combination of these manipulations can be used to generate a library in which each molecule has a distinct identifying sequence." *Id.*

Kivioja's Figure 1(a) is reproduced below.



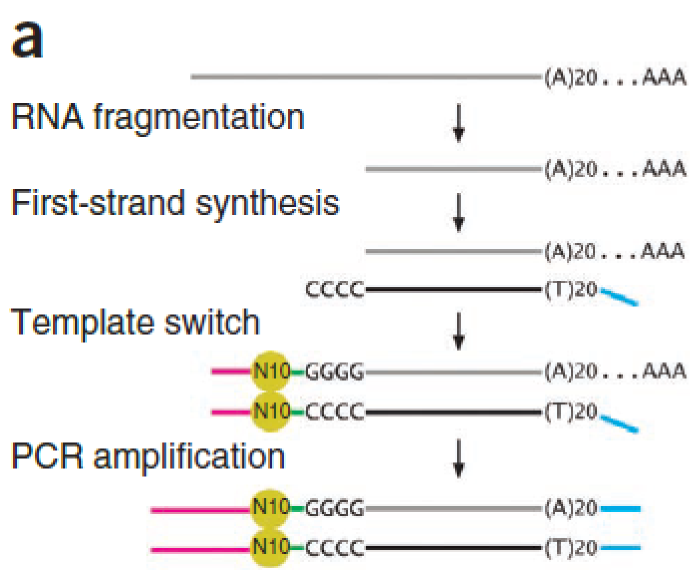
Kivioja describes Figure 1(a) as follows:

> Three different DNA species (green, blue and black lines) are labeled with a collection of random labels (colored filled circles). Two green molecules are originally present (top), corresponding to two different UMIs (red, blue) among the sequenced

> molecules (green; bottom). Information about the original number of molecules (top) is preserved in the number of different UMIs detected by sequencing a sample of the amplified and normalized library (bottom). Even if some UMIs are not observed, the original number of molecules can be estimated using count statistics.

Id.

Kivioja states that a labeling protocol for counting cellular mRNA molecules was also tested. Ex. 1012, 2–3. Specifically, *Drosophila melanogaster* S2 cell RNA was fragmented and converted to cDNA via oligo(dT)-primed reverse transcription and an oligonucleotide with a 10-base-pair random label was incorporated by template switching. *Id.* at 3. This is depicted in Figure 3(a), which is reproduced below.



Referencing Figure 3(a), Kivioja explains that "mRNA-seq libraries were generated by fragmenting total RNA and reverse-transcribing it to cDNA using an oligo(dT) primer with an Illumina linker (blue) and a 5′ template switch adaptor containing another Illumina linker (magenta), a 10-base-pair random label (N10) and an index sequence (green)." *Id.* at 2. Kivioja states that "[t]he combination of label sequence and the 5′ mapped position of the RNA fragment forms the UMI." *Id.*

2. Analysis of Independent Claim 1

a) "A method for detecting duplicate sequencing reads, the method comprising" (preamble)

Petitioner asserts that to the extent the preamble is limiting and entitled to patentable weight, Kivioja teaches this element in the method of collapsing labeled fragments to a single UMI based on the tag and fragment sequence. Pet. 36 (citing, e.g., Ex. 1079 ¶¶ 236–240, Ex. 1012, 1, 5). Petitioner acknowledges that Kivioja does not use the word "duplicate," but argues that Kivioja uses its mRNA sequencing method to detect the number of molecules after different numbers of PCR cycles, that this method necessarily involves the determination of whether particular observed sequences from the Illumina run were original fragments or PCR copies, and that a person of ordinary skill would have recognized that Kivioja teaches a method of detecting and removing duplicates from sequencing information. Pet. 37 (citing Ex. 1079 ¶¶ 258–259).

Patent Owner has not provided separate arguments regarding claim 1's preamble. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that Kivioja discloses that "with the UMI method one can reproducibly detect the number of molecules after different numbers of PCR cycles without normalization." Ex. 1012, 3. We credit the testimony of Dr. Sims, who opines that a person of ordinary skill would have understood that Kivioja's method teaches detecting and removing duplicate sequencing reads, i.e., that Kivioja's method of counting unique sequences necessarily identifies duplicate reads. *See* Ex. 1079 ¶ 260.

b) "ligating an adaptor to each of a plurality of nucleic acid fragments"

Petitioner asserts that Iafrate discloses ligating each sample with unique barcode adaptors. Pet. 26 (citing Ex. 1011, 29). Petitioner concedes that Kivioja's mRNA method uses a reaction called an RNA template switch to attach the adaptor to the 5′ end of the fragments, rather than a ligation reaction, but argues that a person of ordinary skill would have understood that this reaction could only be used with RNA and that Kivioja separately discloses ligating adaptors to DNA. Pet. 31–32 (citing Ex. 1012, 1–3; Ex. 1079 ¶¶ 244–245). Petitioner argues that it would have been obvious to a person of ordinary skill to use the ligation method for an application like Iafrate's. Pet. 32 (citing, e.g., Ex. 1079 ¶¶ 210–211).

Patent Owner has not provided separate arguments regarding this limitation. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that Iafrate discloses that double-stranded cDNA can be sheared, end-repaired, adenylated, and then ligated on one or both ends with an adapter for initiating PCR and sequencing. *See* Ex. 2011 ¶ 230.

c) [wherein each adaptor comprises] "an indexing site unique to a subset of the adaptors"

Petitioner asserts that the Iafrate adaptor includes an indexing site/sample index (referred to as a "barcode"). Pet. 25 (citing Ex. 1079 ¶ 211; Ex. 1011, 11, 13). Petitioner asserts that a person of ordinary skill would have known that Iafrate's sample index would have been unique among a subset of adaptors. *Id.* at 26–27 (citing Ex. 1004, 25:41; Ex. 1079 ¶¶ 213–218).

Patent Owner has not provided separate arguments regarding this limitation. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that Figure 9 of Iafrate discloses an oligonucleotide-tail adapter that includes a barcode. *See* Ex. 1011 ¶ 33, Fig. 9. We credit the testimony of Dr. Sims, who indicates that a person of ordinary skill would have understood a barcode is another term for an indexing site. *See* Ex. 1079 ¶ 211. Dr. Sims appears to rely on general knowledge in the art in interpreting Iafrate, specifically that it generally was known that the sample index allows a particular sequencing read to be associated with its sample of origin (e.g., Sample A or B), where the sample index is unique to each sample and allows sequencing reads to be demultiplexed. *Id.* ¶ 217.

d) [wherein the adaptor comprises] "a unique identifier site having from about 1 to about 8 nucleotides"

Petitioner asserts Kivioja discloses how to enhance the accuracy of Iafrate's method by adding the remaining feature—a random molecular tag—and teaches that random molecular tags are used in combination with a fragment sequence as a "unique molecular identifier" ("UMI") for the identification of duplicate sequencing reads. Pet. 25 (citing Ex. 1012, 1, 5, 14). Petitioner argues that the length of the molecular tag is a paradigmatic result-effective variable. Pet. 39. Petitioner argues that optimizing the length is simply a function of the known relationship between the number of resulting tags (i.e., 4^n) and the expected number of unique fragments in a particular sample. *Id.* (citing, e.g., Ex. 1079 ¶¶ 264–266).

Patent Owner has not provided separate arguments regarding this limitation. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that Kivioja discloses a method of making each molecule in a population unique, including by adding a random DNA sequence label, by fragmenting, or taking an aliquot, such that each molecule has a distinct identifying sequence, and designating the resulting sequences that can be used to uniquely identify copies derived from each molecule as unique molecular identifiers (UMIs). *See* Ex. 1012, 1.

Based on the foregoing, we determine that Petitioner has shown by a preponderance of the evidence that the length of the sequence is a result-effective variable that can be optimized. We credit the testimony of Dr. Sims, who opines that the number of unique molecular tags was a simple function of the length of the molecular tag (4^n), and if the number of target molecules changes, so too does the optimal number of labels required to ensure efficient and unique labeling. *See* Ex. 1079 ¶ 264.

e) [wherein the adaptor comprises] "a primer binding site"

Petitioner asserts that the Iafrate adaptor includes two primer binding sites, i.e., an indexing primer binding site (referred to as a "barcode sequencing primer binding site") and a target sequence binding site (referred to as a "sequencing primer binding site"). Pet. 25–26 (citing Ex. 1079 ¶ 211; Ex. 1011, 11, 13).

Patent Owner has not provided separate arguments regarding this limitation. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that Figure 9 of Iafrate discloses an oligonucleotide-tail adapter that includes a barcode sequencing primer binding site and a sequencing primer binding site. Ex. 1011 ¶ 33, Fig. 9. We credit the testimony of Dr. Sims, who indicates that a person of ordinary skill would have understood a barcode sequencing primer binding site is another term for an indexing primer binding site, and that a sequencing primer binding site is another term for a target sequence binding site. *See* Ex. 1079 ¶ 211.

f) "amplifying the adaptor-ligated fragments into amplicons"; and

g) "sequencing the amplicons to produce sequence reads that include identifier and target sequences;"

Petitioner argues Kivioja discloses amplifying the tagged fragments and sequencing them. *See* Pet. 30, 32, 36 (citing, e.g., Ex. 1012, 1, 4, 5; Ex. 1079 ¶ 247).

Patent Owner has not provided separate arguments regarding these limitations. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that in Kivioja's mRNA experiments, the libraries were amplified using Phusion High-Fidelity DNA polymerase, the PCR products were purified, and then subjected to Illumina GAIIx massively parallel sequencing. *See* Ex. 1012, 4; Ex. 1079 ¶ 247.

h) "identifying sequence reads with identical identifier and target sequences as duplicates"

In discussing the preamble above, we found that Kivioja discloses identifying a duplicate sequencing read. As to "identical identifier and target sequences," Petitioner argues that the combination of the fragment sequences (mapped against known Drosophila genes) and the random label sequence are used to determine whether particular sequence reads are unique or PCR duplicates. Pet. 32 (citing Ex. 1079 ¶ 248).

Patent Owner contends that Kivioja fails to disclose a process of identifying duplicates based on both a portion of the target sequence used in conjunction with a label (an identifier site) in the same step. PO Resp. 25; Sur-Reply 18. Patent Owner acknowledges that Kivioja describes looking to the "gene, position and label," but Patent Owner asserts Kivioja describes a three-step process of counting reads, fails to use both the target sequence and the label in combination in the same, and does not meet the claim limitation that requires a single step.[4] PO Resp. 25; Sur-Reply 1.

Patent Owner asserts in Kivioja's first "step," the sequence reads are mapped to a known genome. PO Resp. 26. According to Patent Owner, Kivioja does not describe looking to the actual target sequence itself, but

[4] During oral argument, counsel for Patent Owner states that "the question of a multi-step process is largely irrelevant to the issues before the Board." Tr. 36:4–5. Counsel appears to make this argument because it contends that the target sequence and the identifier are not used in combination. According to counsel, "Kivioja uses the target sequence in a step to map the reads and generate a different data set that is then used in their analysis. But [Kivioja] only uses the identifier to identify duplicates." *Id.* at 35:25–36:2.

rather teaches the use of the gene position. *Id.* (citing Ex. 2016, 29–30 (109:16–113:09)). Then in the alleged second "step," Kivioja discards reads that did not map against the known reference genome. *Id.* at 26–27 (citing Ex. 1012, 5; Ex. 2016, 29 (110:22–111:16)). Lastly, Patent Owner argues that in Kivioja's third "step," the reads with the same mapped gene and position and label are then "collected to one UMI." *Id.* at 27 (citing Ex. 1012, 5). Therefore, Patent Owner asserts that Kivioja uses the mapped gene and position with the label to count the UMI collections. *Id.* (citing Ex. 2035 ¶¶ 107–116; Ex. 2016, 29 (109:20–112:21)).

Patent Owner contends that Kivioja's reliance on the "gene" (i.e., a proposed chromosomal location of the original genomic sequence), its "position" (i.e., the proposed mapped location of the target sequence to the reference genome) does not meet the challenged claim. PO Resp. 27. Patent Owner argues that Kivioja's use of the gene and position in place of the target sequence transforms one data set into a different data point, which is insufficient. Sur-Reply 18 (citing Ex. 1012, 5). Patent Owner specifically argues that an ordinarily skilled artisan would have "understood that use of the mapped gene and position is functionally distinct from the use of a portion of the target sequence." PO Resp. 28. Patent Owner relies on the testimony of Dr. Bustamante to support its position, because he opines that "[m]apping sequence reads to a portion of the reference genome, rather than to a portion of the target sequence itself, lacks the precision of the claimed method." Ex. 2035 ¶ 57.

Petitioner, however, argues that when Kivioja maps a sequence read to a reference genome it is using a "target sequence" in the same manner as described in the '357 patent. Reply 20 (citing Ex. 1004, 17:39–50; Ex. 1124, 154:1–155:2; Ex. 1079 ¶¶ 248, 274–278, 361–363). According to

Petitioner, the alignment process uses the entire target sequence. *Id.* at 20–21 (citing Ex. 1124, 139:20–24; Ex. 2035 ¶ 56); Tr. 19:4–16 (citing Ex. 1124, 91). Petitioner explains that Figure 1(a) of Kivioja demonstrates a method of identifying and removing duplicates using an identifier and target sequence. Reply 21–22 (citing Ex. 1012, 1). Petitioner also cites to Dr. Bustamante's testimony, when he opines that "the alignment tools compare the nucleotide sequence of each sequencing read to the nucleotide sequences contained in the reference genome . . . ." *Id.* at 21 (citing Ex. 2035 ¶ 56). Therefore, Petitioner argues that the UMI is a combination of label and sequence fragment. Pet. 29 (citing Ex. 1012, 1; Ex. 1079 ¶¶ 238–239).

Based on the entirety of the trial record, we do not agree with Patent Owner's reading of either the claims or the prior art because Patent Owner takes too narrow a view of both. Specifically, based on the plain language of Kivioja, we find it uses a combination of the label and the target sequence as the UMI to determine whether there is duplication. With respect to the mRNA experiments, Kivioja states "[t]he mapped reads with the same gene, position and label were collected to one UMI." Ex. 1012, 5.[5] Therefore, we find that Kivioja's UMI includes the gene, position, and label. All that the claim requires is that the method identifies sequence reads with identical identifier and target sequences as duplicates.

Nevertheless, Patent Owner challenges whether Kivioja relies on the sequence of the label and the sequence of the target fragment, when Kivioja states: "The mapped reads with the same gene, position and label were

[5] With respect to the DNA experiments, Kivioja states "we used the 5′ genomic position of each mapped read as the UMI." Ex. 1012, 4. Kivioja also states that "[t]he sequence of the label and the 5′ mapped position of the fragment together define the UMI." *Id.* at 3.

collected to one UMI." Ex. 1012, 5. When Kivioja states that reads with the "same . . . label" were "collected to one UMI," we understand that to mean that the labels have identical sequences. Kivioja also discloses that "two UMIs were merged if they either had identical positions and one mismatch in the label sequences . . . ." *Id.* This indicates that not only were identical labels considered to be duplicates, but also labels that had one mismatch, i.e., in order to limit the effect of sequencing errors. *See id.* Kivioja relied on the combination of label and nucleic acid fragment to identify duplicates. *See* Ex. 1012, 1 ("We designate the resulting sequences that can be used to uniquely identify copies derived from each molecule unique molecular identifiers (UMIs; Fig. 1)."), 3 ("The sequence of the label and the 5′ mapped position of the fragment together define the UMI."). Because claim 1 contains the transition "comprising," the claim is still satisfied if the category of identified duplicate reads contains identical labels and also contains more.

Additionally, we do not agree with Patent Owner's argument that Kivioja uses the position of the target fragment instead of the sequence of the target fragment, and therefore, does not render the claimed subject matter obvious. Kivioja discloses that "[f]or each gene, the sequence of its longest transcript was used as the reference sequence. Reads were discarded from further analysis if they did not contain the constant sequences expected based on oligonucleotide design, mapped to the wrong strand, [or] had a BWA mapping quality score lower than 20 . . . ." Ex. 1012, 5; *see also id.* ("Total read count of a gene is the number of accepted reads mapped to its reference sequence."). Thus, we find that Kivioja teaches or suggests that the mapped position was based on the gene sequence. *See also* Ex. 1079 ¶¶ 248, 276, 277 ("Mapping reads to the genome necessarily required matching the observed sequence of the nucleic acid fragment to the expected sequence.”), 278.[6]

Patent Owner admits that the identifier and target sequence do not have to be used simultaneously. Tr. 34:23–35:19. Instead, Patent Owner challenges whether Kivioja uses the required elements in combination in the same step. Based on the record before us, we agree with Petitioner’s position especially in view of the open-ended nature of the claims and Patent Owner’s own admission that there is no temporal requirement for the use of the identifier and target sequence. Specifically, we find that Kivioja teaches using the combination of the fragment sequences (mapped against known Drosophila genes) and the random label sequence to determine whether particular sequence reads are unique or PCR duplicates. *See* Ex. 1012, 14. The use of both the mapped sequence (as the target sequence) and the random label (as the identifier) meets the claim limitations. This finding is consistent with the prosecution history of the ’357 patent and how the applicant and examiner interpreted the claims. *See* Ex. 1019, 452–53; Ex. 1018, 780–81, 797–99, 807; Ex. 1022, 4–6; Ex. 1033, 86–87, 94, 97, 99.

We, therefore, determine that Petitioner has demonstrated by a preponderance of the evidence that the combined teachings of Kivioja and Iafrate render obvious the claim limitation of “identifying sequence reads with identical identifier and target sequences as duplicates.”

i) Rationale to Combine and Reasonable Expectation of Success

Petitioner contends that a person of ordinary skill would have been motivated to incorporate the Kivioja molecular tag into the Iafrate adaptor in

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[6] Dr. Sims noted that in Kivioja, after sequencing, reads were mapped to the genome. Ex. 1079 ¶ 277 (citing Ex. 1012, 4).

order to realize the Iafrate's method's full potential, i.e., by mitigating PCR errors and improving accuracy. *See* Pet. 33 (citing Ex. 1079 ¶¶ 225–249).

Patent Owner argues that Petitioner's argued motivation is conclusory, based on improper hindsight, or based on the ipse dixit of an expert witness. PO Resp. 31–32. Patent Owner argues that Petitioner glosses over the differences between Iafrate and Kivioja's adaptors on the one hand, and the claimed adapters, on the other. *Id.* at 32. Patent Owner attempts to rebuke Petitioner's argument that Kivioja's target sequence primer binding site serves the purpose of both of the claimed primer binding sites, because Petitioner provides no explanation on why or how an ordinarily skilled artisan would have combined different aspects of the Iafrate and Kivioja adapters. *Id.* Patent Owner then argues that Petitioner fails to explain which aspects of the Iafrate method a person of ordinary skill in the art would have used in any improved sequencing method because Petitioner's explanation of its obviousness arguments focuses solely on the disclosures of Kivioja. *Id.* Patent Owner further argues that Iafrate does not disclose a molecular tag, so Petitioner cannot rely on Iafrate for the disclosure of that limitation. *Id.*

Petition does in certain places rely on both Iafrate and Kivioja for the same limitation, however, we understand the Petition primarily relies on Iafrate for "ligating an adaptor to each of a plurality of nucleic acid fragments, wherein each adaptor comprises . . . an indexing site unique to a subset of the adaptors, and a primer binding site." *See* Pet. 25–26. The Petition is clear, though, that it relies on Kivioja for the random molecular tag and identifier site. *See id.* at 38–39.

Patent Owner argues Iafrate does not provide a suggestion to use a shorter molecular tag and does not use a molecular tag at all. PO Resp. 32–

33 (citing Pet. 38). Patent Owner appears to refer to the following argument in the Petition: "[a]lthough Kivioja used a 10-nucleotide molecular tag, Kivioja provided a specific motivation to use a shorter molecular tag, particularly for an application like Iafrate's." *See* Pet. 38. Petitioner also argues, however, that the length of the molecular tag is a result-effective variable. *See id.* at 39. In context, Petitioner appears to be arguing that a shorter molecular tag would have worked even in Iafrate's method because there would have been a relatively small number of original fragments of interest. *See id.* at 38–39.

Dr. Sims opines that "[t]he POSA would have understood that for the targeted enrichment methods of Iafrate, particularly when combined with a relatively low sample concentration, the ten-nucleotide molecular tag used in Kivioja, which was excessive even for Kivioja's shotgun sequencing method, would have been optimized to a lower number, including a molecular tag with between 1 and 8 nucleotides." Ex. 1079 ¶ 269. Based on our reading of the prior art and the evidence of record, we find Dr. Sims's testimony consistent with the teaching in Iafrate that the amount of starting nucleic acid material may vary. *See* Ex. 1011 ¶ 230; Ex. 1079 ¶ 269 (citing Ex. 1011, 29).

On the complete trial record, we determine that Petitioner has provided a rationale to combine the teachings of the prior art, not based on hindsight, e.g., when Petitioner points to the statement in Kivioja that use of Kivioja's method "can improve accuracy of almost any next-generation sequencing method." Pet. 3, 29 (citing Ex. 1012, 1); Ex. 1012, 1. Therefore, we conclude that Petitioner has demonstrated by a preponderance of the evidence that a person of ordinary skill would have had sufficient reasons to combine the teachings of Iafrate and Kivioja to arrive at the method of claim 1, and would have had a reasonable expectation of success.

j) Summary of Claim 1

On the complete trial record before us now, we determine Petitioner has demonstrated by a preponderance of the evidence that claim 1 of the ’357 patent would have been obvious over Iafrate and Kivioja.

3. Analysis of Dependent Claims 2–4, 6, 7, 9, and 10

Petitioner provides arguments and evidence regarding dependent claims 2–4, 6, 7, 9, and 10 in view of Iafrate and Kivioja. Pet. 41–47.

Other than its arguments directed to Petitioner’s challenges to independent claim 1, Patent Owner does not specifically respond to Petitioner’s arguments regarding dependent claims 3, 4, 6, 7, 9, and 10. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378. Patent Owner does address dependent claim 2.

Claim 2 depends from claim 1 and “further compris[es] removing a duplicate read from the sequence reads.” *See* Ex. 1001, 25:48–49.

Patent Owner contends Petitioner admits that Kivioja fails to disclose this limitation. PO Resp. 30 (citing Paper 1, 41). Patent Owner argues that Petitioner attempts to “cure this defect” by arguing that removing duplicates is inherent in Kivioja. *Id.* But Patent Owner asserts that Petitioner’s interpretation is not supported by Kivioja, and that Petitioner’s expert does not dispute that the UMIs are not removed prior to the counting taught in Kivioja. *Id.* (citing Ex. 2016, 29-30 (111:17-115:1)).

Petitioner disputes Patent Owner’s interpretation of Kivioja and cites to Kivioja’s clear disclosure that duplicate reads are “collected to one UMI” and treated as a single original fragment. Pet. 41 (citing Ex. 1012, 5).

Petitioner further argues that an ordinarily skilled artisan "would have understood that Kivioja's method of counting the original number of molecules in a sample requires deduplication as a predicate step to counting the original fragments." *Id.* (citing Ex. 1079 ¶¶ 364–366, 284–285).

Petitioner also relies on Figures 3b and 3c to support its position that there is a reduction is sequence read through the use of Kivioja's UMI method. Reply 26–27 (citing Ex. 1012, 2). Petitioner explains that Figure 3b depicts post sequencing read counts for a set genes [sic], while Figure 3c depicts the same data after performing deduplication with Kivoja's UMI method. *Id.* at 26 (citing EX1012, 2). According to Petitioner, a comparison of the number of reads, and scales on the x- and y-axes in both Figures, shows there is a roughly 10-fold reduction in sequences using the UMI method of deduplication. *Id.* at 26–27 (citing Ex. 10012, 2). Petitioner then contends that Kivioja illustrates similar data in Supplementary Figure 4. *Id.* at 27 (citing EX1012, 10). Therefore, Petitioner concludes that Kivioja discloses the removal of duplicate sequencing reads from a population of identified duplicates. *Id.*

Patent Owner disagrees with Petitioner's characterization of Figures 3b and 3c. Sur-Reply 20–21. Patent Owner argues that the different values shown in the figures do not provide any information about whether duplicates were removed because the figures are comparing different metrics. *Id.* According to Patent Owner, the "fact that the reads within each of the UMI families are not plotted in this comparison does not mean this data has been removed." *Id.* at 21. Patent Owner then asserts that the data reported in Figure 3b shows all the reads are kept because the total reads are reported. *Id.* (citing Ex. 1012, 2–3).

We disagree with Patent Owner's interpretation of Kivioja, specifically Figures 3b and 3c, and we disagree that Kivioja does not meet the claim limitation. Claim 2 requires "removing a duplicate read from the sequence reads." Based on the disclosure of the '357 patent, that does not mean the actual oligonucleotide sequence must be removed from the library or that the actual oligonucleotide must be destroyed. As the '357 patent makes clear, "removing" a duplicate read from a sequence read simply means that the data associated with the nucleic acid fragment in a computer file must be separated in the file before analysis. *See* Ex. 1004, 15:18–22, 20:60–64 ("Any read that has identical sequence to any other read is identified as a duplicate and removed from the population, thus, only a single copy is retained within the file."). The '357 patent uses

> Standard PCR duplicate removal software such as PICARD, Markduplicates, and/or SAMtools mdup is used to identify and remove the PCR duplicates, leaving behind for analysis any instances of multiple cDNA fragments that happen to have the same sequence.

Id. at 18:66–19:4; 19:31–36. This appears to be exactly what Kivioja does when it collects to one UMI all the sequence reads that map to the same gene, position, and label. *See* Ex. 1012, 5.

For dependent claims 3, 4, 6, 7, 9, and 10, Patent Owner appears to rely on its previously discussed arguments regarding Kivioja and Iafrate, because it has not provided separate arguments regarding these claims. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

We have reviewed the entirety of the trial record and all evidence submitted for claims 2–4, 6, 7, 9, and 10. Based on the complete record, we determine that Petitioner has shown that Kivioja in combination with Iafrate teaches or suggests the limitations as recited in dependent claims 2–4, 6, 7, 9, and 10. Accordingly, we determine that Petitioner has established by a preponderance of the evidence that claims 2–4, 6, 7, 9, and 10 are unpatentable under 35 U.S.C. § 103 over Iafrate and Kivioja.

B. Alleged Anticipation of Claims 1–4, 6, 7, and 10 by Bielas

Petitioner contends that claims 1–4, 6, 7, and 10 are anticipated by Bielas. Pet. 47–65; Reply 14–19. Patent Owner disagrees. PO Resp. 33–38; Sur-Reply 12–16.

1. Bielas

Bielas is a Patent Cooperation Treaty publication of an international application titled "Compositions and Methods for Accurately Identifying Mutations." Ex. 1013, code (54). Bielas "relates to compositions and methods for accurately detecting mutations using sequencing and, more particularly, uniquely tagging double stranded nucleic acid molecules such that sequence data obtained for a sense strand can be linked to sequence data obtained from the anti-sense strand when obtained via massively parallel sequencing methods." *Id.* at 1:15–19.[7]

Bielas explains that the detection of spontaneous mutations and induced mutations that occur randomly throughout a genome can be difficult because such mutations "are rare and may exist in one or only a few copies of DNA." Ex. 1013, 1:21–24. According to Bielas, however, "a mutation must be present in as much as 10–25% of a population of molecules to be detected by sequencing." *Id.* at 1:25–2:3. To address this, Bielas describes

> a double-stranded nucleic acid molecule library that includes a plurality of target nucleic acid molecules and a plurality of

[7] For Bielas, we use the native reference page numbers as opposed to the page numbers added to the exhibit.

> random cyphers, wherein the nucleic acid library comprises molecules having a formula of X^a-X^b-Y, X^b-X^a-Y, Y-X^a-X^b, Y-X^b-X^a, X^a-Y-X^b, or X^b-Y-X^a (in 5′ to 3′ order), wherein (a) X^a comprises a first random cypher, (b) Y comprises a target nucleic acid molecule, and (c) X^b comprises a second random cypher. Furthermore, each of the plurality of random cyphers comprise a length ranging from about 5 nucleotides to about 50 nucleotides (or about 5 nucleotides to about 10 nucleotides, or a length of about 6, about 7, about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15, about 16, about 17, about 18, about 19, or about 20 nucleotides).

Id. at 2:29–3:8.

Bielas describes a first example of dual cypher sequencing of normal and tumor genomic libraries. *See* Ex. 1013, 24:12–15. Bielas states that isolated genomic DNA is fragmented by shearing and "[t]he DNA fragments having overhang ends are repaired (*i.e.,* blunted) using T4 DNA polymerase (having both 3′ to 5′ exonuclease activity and 5′ to 3′ polymerase activity) and the 5′-ends of the blunted DNA are phosphorylated with T4 polynucleotide kinase . . . , and then purified." *Id.* at 24:17–22. Furthermore, "[t]he end-repaired DNA fragments are ligated into the *Sma*I site of the library of dual cypher vectors shown in Figure 2 to generate a target genomic library." *Id.* at 24:23–24.

Bielas states that this *Sma*I site is shown in Figure 2, which is reproduced below. Ex. 1013, 24:23–24.

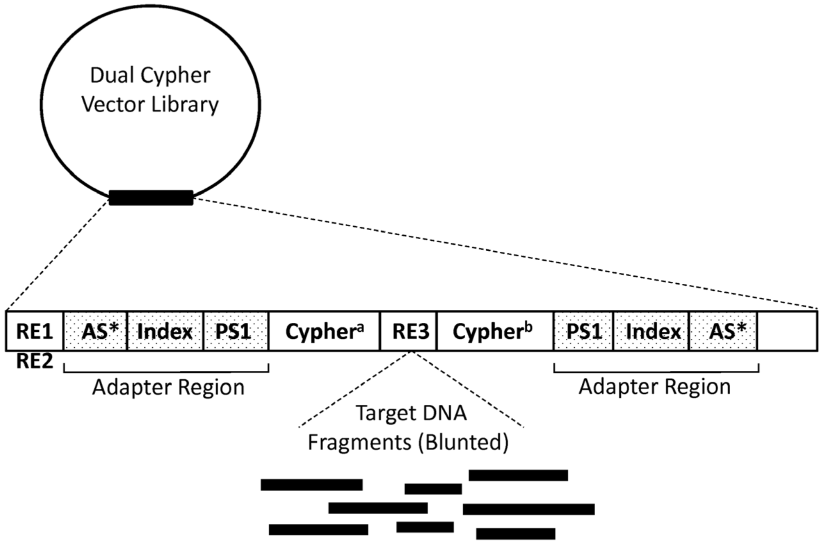

Fig. 2

Figure 2 depicts an exemplary vector, "wherein adaptor sequences are included and are useful for, for example, bridge amplification methods before sequencing." *Id.* at 4:1–3. Bielas explains that "[t]he ligated cypher vector library is purified and the target genomic library fragments are amplified" via a PCR protocol. *Id.* at 24:25–28. Bielas further states:

> The amplification is performed using sense strand and anti-sense strand primers that anneal to a sequence located within the adapter region (in certain embodiments, the primer will anneal to a sequence upstream of the AS), and is upstream of the unique cypher and the target genomic insert (and, if present, upstream of an index sequence if multiplex sequencing is desired; *see, e.g.,* Fig. 2) for Illumina bridge sequencing.

Id. at 24:28–25:4. Bielas states that "[t]he unique cypher tags are used to computationally deconvolute the sequencing data and map all sequence reads to single molecules (*i.e.,* distinguish PCR and sequencing errors from real mutations)." *Id.* at 25:7–9.

Bielas describes another example where "[d]ual cypher sequencing of present disclosure was used to detect somatic TP53 mutations that arose during replication in *E. coli.*" *Id.* at 26:21–22. When explaining results, Bielas states:

> TP53 Exon 4 DNA from a dual cypher vector library produced in *E. coli* was sequenced with a depth of over a million, and all sequencing reads with identical cypher pairs and their reverse complements were grouped into families to create a consensus sequence. As illustrated in Figure 3B, errors introduced during library preparation (open circle) and during sequencing (gray circle and triangle) were computationally eliminated from the consensus sequence and only mutations present in all reads (black diamonds, Figure 3B) of a cypher family were counted as true mutations (*see* bottom of Figure 3B).

Id. at 29:26–30:2. Figure 3B is reproduced below.

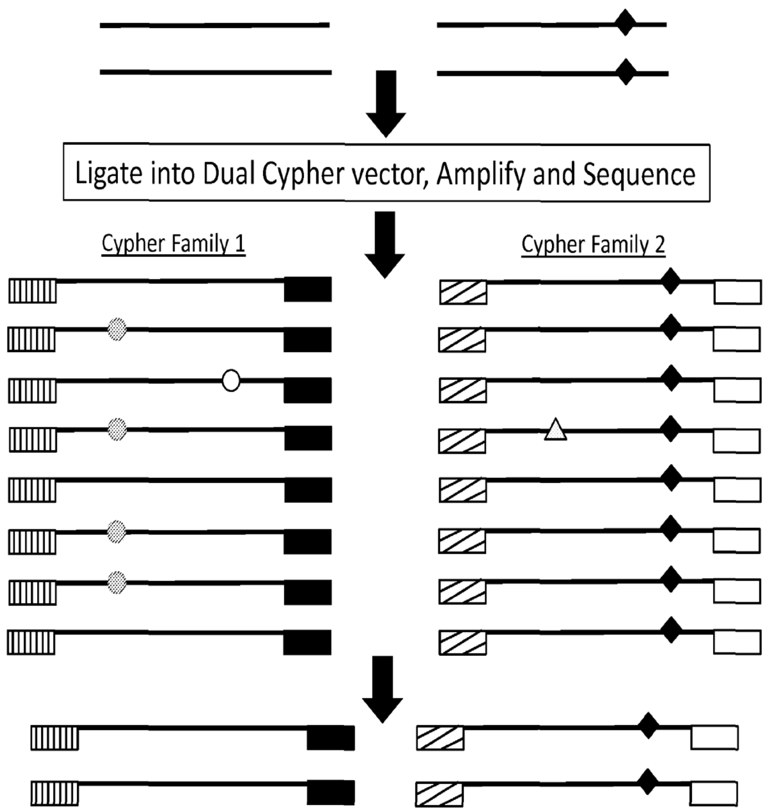

Fig. 3B

2. *Analysis of Independent Claim 1*

Petitioner asserts that Bielas discloses all of the limitations of claim 1. Pet. 53–63; Reply 14–19. Patent Owner argues that Bielas does not disclose identifying sequence reads with identical identifier and target sequences as duplicates. *See* PO Resp. 33–38; Sur-Reply 12–16.

a) *"A method for detecting duplicate sequencing reads, the method comprising" (preamble)*

Petitioner contends Bielas discloses a method for detecting duplicate sequencing reads. Pet. 54 (citing Ex. 1079 ¶¶ 489–491, 566–567). Petitioner asserts that "[d]espite not using the word 'duplicate,' Bielas teaches differentiating between 'true mutations' and 'artifact mutations' using the combination of the Cypher/identifier site and the fragment so that each sequence 'can be connected back to the original molecule.'" *Id.* (citing Ex. 1013, 7; Ex. 1079 ¶¶ 489–491, 566–567).

Patent Owner has not provided separate arguments regarding claim 1's preamble. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that Bielas discloses a "method for detecting duplicate sequencing reads" in its Examples 1 and 3. *See* Ex. 1013, 7, 24–25, 28–29, 31–32. We credit the testimony of Dr. Sims, who explains how Bielas uses the Cypher and "sequence alignment" to the reference genome to "computationally deconvolute" the data to "distinguish PCR and sequencing errors from real mutations." *See* Ex. 1079 ¶ 490.

b) "ligating an adaptor to each of a plurality of nucleic acid fragments"

Petitioner asserts that Bielas discloses ligation of a fragment into a circular vector, which attached an adaptor made up of AS-Index-PS1-Cypher to the 5′ end of the fragment. Pet. 55 (citing Ex. 1013, 26, 29–31, Fig. 3; Ex. 1079 ¶¶ 492–505, 568–572).

Patent Owner has not provided separate arguments regarding this limitation of claim 1. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that Examples 1 and 3 of Bielas disclose that double-stranded cDNA can be fragmented with overhanging ends that are blunted and then ligated into the *Sma*I site of a dual cypher vector to generate a target genomic library. *See* Ex. 1013, 26, 29–31, Fig. 3.

c) [wherein each adaptor comprises] "an indexing site unique to a subset of the adaptors"

Petitioner asserts that Bielas's adaptor includes a sample index/indexing site in the 5′ portion of the adaptor. Pet. 57 (citing Ex. 1013, 27–29, 41; Ex. 1079 ¶¶ 495–497, 510, 569). Petitioner cites to Bielas's Example 1 and the adaptor disclosure there because it contains "an index sequence if multiplex sequencing is desired." *Id.* at 58 (citing Ex. 1013, 27). In addition, Petitioner argues that in Figure 2, the adaptor structure is shown and it "includes an Index site, defined as a sample index, in the 5′ side." *Id.* (citing Ex. 1013, 27, 41, Fig. 2; Ex. 1079 ¶ 495). Petitioner also relies on Bielas's Example 3 and the adaptor disclosed there because it "includes at the 5′ side a standard Illumina sample index." *Id.* (citing Ex. 1013, 28–29;

Ex. 1079 ¶¶ 496–497). According to Petitioner, "[t]his is consistent with Bielas's Figure 2 and other teachings that Bielas included sample indexes in the adaptors." *Id.* (citing Ex. 1079 ¶¶ 495–497). Petitioner next argues that Bielas explains that these indexes are "a unique index sequence . . . specific for a particular sample," and that if ten samples are pooled, ten different index sequences will be used. *Id.* (citing Ex. 1013, 15–16; Ex. 1079 ¶ 510).

Patent Owner has not provided separate arguments regarding this limitation of claim 1. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that Examples 1 and 3 as well as Figure 2 of Bielas discloses an index sequence. *See* Ex. 1013, 15–16, 27–29, 41.

d) [wherein the adaptor comprises] "a unique identifier site having from about 1 to about 8 nucleotides"

Petitioner contends Bielas discloses that the Cypher sequence in some embodiments consists of specifically 6, 7, or 8 nucleotides. Pet. 56 (citing Ex. 1013, 15; Ex. 1079 ¶¶ 499–504). Additionally, Petitioner argues that Bielas states that, in discussing a similar construct to that shown in Figure 2, the Cypher is "preferably from about 7 nucleotides to about 9 nucleotides." *Id.* (citing Ex. 1013, 23; Ex. 1079 ¶ 503). Petitioner further asserts that in Example 3, Bielas discloses a 7-nucleotide random Cypher in the 5′ portion of the adaptor, designated as NNNNNNN. *Id.* at 57 (citing Ex. 1013, 28; Ex. 1079 ¶ 504). According to Petitioner, the disclosure of an adaptor with a 7-nucleotide identifier site that falls within the range of "about 1 to about 8" anticipates this claim limitation. *Id.*

Patent Owner has not provided separate arguments regarding this limitation of claim 1. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that Bielas discloses an adaptor with a 7-nucleotide identifier site that falls within the range of "about 1 to about 8." *See* Ex. 1013, 15, 23, 28.

e) [wherein the adaptor comprises] "a primer binding site"

Petitioner contends that Bielas discloses a sequencing primer binding site/target sequence primer binding site in both Example 1 and Example 3. Pet. 58 (citing Ex. 1013, 27–29; Ex. 1079 ¶ 505). Petitioner argues that "[i]n Example 1, Bielas used the adaptor in Figure 2 and the adaptor-fragments were sequenced on an Illumina device." *Id.* at 58–59 (citing Ex. 1013, 27). Petitioner asserts that Figure 2 discloses that the adaptor included "PS," which Petitioner notes is defined by Bielas as "a nucleic acid molecule priming site" "used to initiate a sequencing reaction." *Id.* at 59 (citing Ex. 1013, 12). Based on the combination of these teachings, Petitioner argues that a person of ordinary skill in the art would have understood that PS was a primer binding site for an Illumina sequencing primer. *Id.* (citing Ex. 1013, 41; Ex. 1079 ¶¶ 466–467, 505).

Petitioner also asserts that in Example 3, the 5′ adaptor sequence disclosed by Bielas includes an indexing primer binding site, which begins at the AS sequence and ends immediately before the Index. *Id.* at 60 (citing Ex. 1079 ¶¶ 474–475). According to Petitioner, the AS sequence (also the standard Illumina P5 sequence for annealing to the solid support) functions as the indexing primer binding site on an Illumina MiSeq, which Bielas used in Example 3. *Id.* at 60 (citing Ex. 1013, 28–29; Ex. 1079 ¶ 474).

Patent Owner has not provided separate arguments regarding this limitation of claim 1. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that Bielas discloses a sequencing primer binding site/target sequence primer binding site in both Example 1 and Example 3. *See* Ex. 1013, 12, 27–29, 41.

f) "amplifying the adaptor-ligated fragments into amplicons"; and

g) "sequencing the amplicons to produce sequence reads that include identifier and target sequences;"

Petitioner contends in Examples 1 and 3, Bielas amplified adaptor-ligated fragments into amplicons using a PCR reaction with primers complementary to the AS sequences on either side of the adaptor to generate copies of the adapter-ligated fragments. Pet. 60 (citing Ex. 1013, 26–27, 31; Ex. 1079 ¶¶ 506, 573–576). Following amplification, Petitioner argues that Bielas sequenced the amplicons to produce sequence reads in a single run of an Illumina device. *Id.* (citing Ex. 1013, 27, 31; Ex.1079 ¶¶ 507, 575).

Patent Owner has not provided separate arguments regarding these limitations of claim 1. *See generally* PO Resp. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

After review of the entire trial record, we agree with Petitioner's position and find that in Bielas discloses an amplifying ligated fragments into amplicons using a PCR and sequencings those amplicons in both Example 1 and Example 3. *See* Ex. 1013, 26–27, 31.

h) "identifying sequence reads with identical identifier and target sequences as duplicates"

Petitioner asserts that Bielas discloses that fragments can be "computationally deconvolute[d]" using cypher-fragment sequences to distinguish "true mutations" found in original fragments from "artifact 'mutations'" generated during library preparation. Pet. 60–61 (citing Ex. 1013, 24–25, 27, 37, Fig. 1). Petitioner argues that regardless of the "objective" of Bielas, this process requires Bielas to first identify duplicate sequencing reads as those comprising the same Cypher/identifier site and fragment sequence as other reads. *Id.* at 61 (citing Ex. 1079 ¶¶ 577–579, 508–509); Reply 15.

Petitioner relies on Bielas's Figure 3 to support its position. Petitioner argues that the original double stranded fragments of Figure 3 are ligated into adaptors containing Cyphers/identifier sequences. Pet. 62 (citing Ex. 1013, 43; Ex. 1079 ¶¶ 483–486). According to Petitioner, the fragments are then amplified, creating multiple different copies identical to the original fragment. *Id.* Petitioner contends that in Bielas's Example 3, "all sequencing reads with identical cypher pairs and their reverse complements were grouped into families to create a consensus sequence" and then "errors introduced during library preparation and during sequencing were computationally eliminated from the consensus sequence and only mutations present in all reads of a cypher family were counted as true mutations." Pet. 61–62 (citing Ex. 1013, 31–32). Petitioner further argues that, to perform this comparison, Bielas looks at both the cypher sequence, to group reads, and the fragment sequence, to ascertain whether a particular sequence contained a mutation. *Id.* at 61–62 (citing Ex. 1013, 31–32, 43; Ex. 1079 ¶¶ 483–486). Petitioner then explains that Bielas de-duplicates these reads with identical cypher and fragment sequences to a single consensus read in order to compare those reads to the reference genome and identify "true mutations." *Id.* at 62.

Petitioner argues that the challenged claims do not require the identifier and target sequence be used in a single "step" as asserted by Patent Owner. Reply 15. According to Petitioner, Bielas's method in Example 3 obtains sequence reads with identifiers and target sequences, sorts those sequence reads by comparing identifiers, and then compares target sequences. *Id.* (citing Ex. 1079 ¶¶ 483–486; Ex. 1124, 163:5–166:24; 172:3–172:20). Petitioner concludes that even if the process is done "in series," that still meets the claim limitation. *Id.*

Patent Owner argues that neither Bielas nor Petitioner provides an explanation as to what "deconvoluting" in Bielas entails, and even if deconvoluting identifies duplicates, Bielas does not disclose the use of both the identifier site and any part of a nucleic acid fragment to identify duplicates as part of this step. PO Resp. 34–35; Sur-Reply 13. According to Patent Owner, "[a]ny finding that deconvolute equates to identifying duplicates would be based on pure speculation." Sur-Reply 13.

Patent Owner further argues that Bielas cannot meet the claims because (1) it breaks its process into series of steps to analyze sequence reads and (2) "Example 1 does not teach that both the cyphers ***and*** the portion of the target sequence were used to 'deconvolute' the data." PO Resp. 35. Patent Owner asserts that Bielas merely states that "unique cypher tags" were used to "computationally deconvolute" the data, without any mention of, or reference to, the target sequence. *Id.* (citing Ex. 1013, 25);

Sur-Reply 13 (citing Ex. 1013, 27). Patent Owner argues that Petitioner's declarant, Dr. Sims, only provides unsupported testimony that should be rejected. PO Resp. 35.

Patent Owner also argues that Example 3 of Bielas does not support Petitioner's position. *Id.* Patent Owner contends sequence reads are only grouped together based on Cypher pairs without reference to the target sequence. *Id.* at 35–36 (citing Ex. 1013, Fig. 3; Ex. 2016, 33–34 (128:18–129:15) (explaining that in Example 3 of Bielas, "[t]he cypher pairs and their reverse complements are used to group sequencing reads into families"). Therefore, Patent Owner concludes that "Bielas does not describe using the target sequence to create the cypher families."

Patent Owner then asserts that the Bielas method also fails to teach a method of identifying duplicates, as described in the claims. Patent Owner argues that the purpose of Bielas is to determine if a mutation is an artifact of PCR or a true mutation in the sample. PO Resp. 36 (citing Ex. 2016, 32 (121:6–122:7)). According to Patent Owner, Bielas does not disclose that "the fragments are used to identify duplicates under the proper construction of the challenged claims; rather, Bielas's method allows a POSA to more accurately distinguish true mutations . . . from artifact 'mutations.'" *Id.* at 37 (citing Ex. 1013, 7; Ex. 2025 ¶¶ 117–124, 143, 144). Patent Owner further argues that this "is precisely the type of method that Applicants distinguished from the claimed method" during prosecution. PO Resp. 37; Sur-Reply 14 (citing Ex. 1033, 98–99).

We do not agree with Patent Owner. As explained previously, claim 1 contains the transition "comprising," therefore the claim is still satisfied if the category of identified duplicate reads contains identical nucleic acid fragments and also contains more. Bielas identifies apparent duplicate sequencing reads by first comparing the Cypher/identifier site and grouping them into families to create a "consensus sequence." Ex. 1013, 25:7–8, 31–32. Then Bielas sequences the fragment and compares the actual sequence of the fragment to the consensus sequence. Any sequence with an "error" or mutation is removed from the consensus sequence. That way, at the end, Bielas has all sequences reads with the same Cypher/identifier and the same sequence grouped together in a single consensus read. Ex. 1013, 31–32, 43.

Patent Owner admitted that the identifier and target sequence do not have to be used simultaneously. Tr. 34:23–35:19. Instead, Patent Owner challenges whether Bielas uses the required elements in combination in the same step and for the same "objective." Based on the record before us, we agree with Petitioner's position especially in view of the open-ended nature of the claims and Patent Owner's own admission that there is no temporal requirement for the use of the identifier and target sequence.

Bielas differentiates true mutations from mutations created during library preparation. *See* Ex. 1013, 29–30. We specifically find that Bielas discloses using the combination of the Cypher/identifier site and fragment sequences to determine whether particular sequence reads are unique (or "true mutations") or PCR duplicates. *Id.* at 29:23–24, 30–32, 43. The use of both the fragment sequence (as the target sequence) and the Cypher (as the identifier) meet the claim limitations regardless of objective or if they are not done at the same time.

We also find that Bielas states that "[b]arcodes were used to deconvolute the sequencing data." *See* Ex. 1013, 29:23–24. Bielas also states that "[t]he unique cypher tags are used to computationally deconvolute the sequencing data and map all sequence reads to single molecules." *Id.* at 25:7–8. Based on this language and the usage in Bielas, we agree with

Petitioner an ordinarily skilled artisan would have understood that "deconvolute" means to "deduplicate." Although Dr. Simms may have used this term differently in his own writings, his testimony regarding its usage and meaning in Bielas is credible because it is consistent with Bielas. *See* Ex. 1079 ¶ 470. Thus, we find Bielas uses the cyphers to perform deconvolution and deduplicate the sequences. *See* Ex. 1013, 25, 29.

This finding is consistent with the prosecution history of the '357 patent and how the applicant and examiner interpreted the claims. *See* Ex. 1019, 452–53; Ex. 1018, 780–81, 797–99, 807; Ex. 1022, 4–6; Ex. 1033, 86–87, 94, 97, 99. We do not agree with Patent Owner that Bielas is similar to Samuels (a reference cited during prosecution) because Samuels only identifies differences through comparing barcodes/identifiers, and is distinct from the method claimed. *See* Ex. 1019, 452. Rather, as previously explained, Bielas also uses the fragment sequence to compare and identify duplicates. Even if Bielas's objective "is to determine if a mutation is an artifact of PCR or a true mutation in the sample" it still performs the claimed limitations. *See* Reply 15.

We, therefore, determine that Petitioner has demonstrated by a preponderance of the evidence that the combined teaching of Bielas anticipates the claim limitation of "identifying sequence reads with identical identifier and target sequences as duplicates."

i) Summary of Claim 1

On the complete trial record before us now, we determine Petitioner has demonstrated by a preponderance of the evidence that claim 1 of the '357 patent is anticipated by Bielas.

3. Analysis of Dependent Claims 2–4, 6, 7, and 10

Petitioner provides arguments and evidence regarding anticipation of claims 2–4, 6, 7, and 10 based on Bielas. Pet. 63–65. Patent Owner appears to rely on its previously discussed arguments regarding Bielas and claim 1, because it has not provided separate arguments regarding these claims. *See* PO Resp. 33–37; Sur-Reply 12–16. Nonetheless, the burden remains on Petitioner to demonstrate unpatentability. *See Dynamic Drinkware*, 800 F.3d at 1378.

We have reviewed the entirety of the trial record and all evidence submitted for claims 2–4, 6, 7, and 10. Based on the complete record, we determine that Petitioner has established by a preponderance of the evidence that Bielas discloses the limitations as recited in claims 2–4, 6, 7, and 10.

C. Alleged Obviousness of Claims 1–4, 6, 7, 9, and 10 over Bielas

Petitioner contends that claims 1–4, 6, 7, 9, and 10 would have been obvious over Bielas. Pet. 66–70. Patent Owner disagrees. PO Resp. 37-38. Patent Owner specifically argues that Petitioner fails to provide any proposed modification of Bielas that would arrive at the challenged claims. *Id.* at 38. Therefore, Patent Owner concludes that Petitioner's obviousness argument cannot cure the deficiencies of Bielas. *Id.*

We disagree with Patent Owner. As our reviewing court has stated, "it is well settled that a disclosure that anticipates under § 102 also renders the claim invalid under §103, for 'anticipation is the epitome of obviousness.'" *See Realtime Data, LLC, v. Iancu*, 912 F.3d 1368, 1373 (Fed. Cir. 2019) (citing *Connell v. Sears, Roebuck & Co.*, 722 F.2d 1542, 1548 (Fed. Cir. 1983) (quoting *In re Fracalossi*, 681 F.2d 792, 794 (CCPA 1982)). We find that Bielas discloses every limitation of the challenged claims and would have rendered such limitations obvious to a person of ordinary skill in the art. We credit the testimony of Dr. Sims, who explains how Bielas could be used with standard sequencing Illumina devices, indexes, and adaptor sequences with a reasonable expectation of success. *See* Ex. 1079 ¶ 580, ¶¶ 512–519, ¶¶ 456–486. Dr. Bustamante does not explain how or why the Bielas method could not be used in such a way.

Accordingly, we determine that Petitioner has established by a preponderance of the evidence that claims 1–4, 6, 7, 9, and 10 are unpatentable under 35 U.S.C. § 103 over Bielas.

IV. CONCLUSION[8]

For the foregoing reasons, we are persuaded that Petitioner establishes by a preponderance of the evidence that claims 1–4, 6, 7, 9, 10 of the '357 patent have been shown to be unpatentable.

In summary:

Claims	**35 U.S.C. §**	**References/ Basis**	**Claims Shown Unpatentable**	**Claims Not Shown Unpatentable**
1–4, 6, 7, 9, 10	103	Iafrate, Kivioja	1–4, 6, 7, 9, 10	
1–4, 6, 7, 10	102	Bielas	1–4, 6, 7, 10	
1–4, 6, 7, 9, 10	103	Bielas	1–4, 6, 7, 9, 10	

[8] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

Claims	35 U.S.C. §	References/ Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
Overall Outcome			1–4, 6, 7, 9, 10	

V. ORDER

In consideration of the foregoing, it is hereby

ORDERED that claims 1–4, 6, 7, 9, 10 of U.S. Patent No. 11,098,357 B2 have been shown to be unpatentable under 35 U.S.C. § 102 and § 103; and

FURTHER ORDERED that, because this is a final written decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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